

Literature review evidence table — sources R01-R02 (1/9)

| Reference                                                                                                                                    | Main contribution / evidence                                             | Role in thesis                                                                    | Source and citation status                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|
| R01   Workload foundations<br>Vaswani et al. (2017). Attention Is All You Need. NeurIPS.                                                     | Defines the Transformer and scaled dot-product self-attention.           | Provides the attention workload foundation; report prefill and decode separately. | Peer-reviewed. Found in DOCX and PPTX.                               |
| R02   Workload foundations<br>Devlin et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL. | Shows the Transformer as a reusable large-scale language-model workload. | Justifies realistic Transformer inference evaluation and model selection.         | Peer-reviewed. PPTX only; citation completed from standard metadata. |

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R03-R04 (2/9)

| Reference                                                                                                                                                          | Main contribution / evidence                                                     | Role in thesis                                                                       | Source and citation status    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------|
| R03   Workload foundations<br>Sze et al. (2017). Efficient Processing of Deep Neural Networks: A Tutorial and Survey. Proceedings of the IEEE, 105(12), 2295–2329. | Surveys data reuse, energy, throughput, hardware, and evaluation trade-offs.     | Supports measuring memory, latency, and throughput together rather than FLOPs alone. | Peer-reviewed. DOCX and PPTX. |
| R04   Platforms and profiling<br>Shen et al. (2018). CUDAAAdvisor: LLVM-Based Runtime Profiling for Modern GPUs. CGO, 214–227.                                     | Profiles runtime and memory behavior across CUDA versions and GPU architectures. | Use profiler evidence to attribute a speedup; do not rely on elapsed time alone.     | Peer-reviewed. DOCX and PPTX. |

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R05-R06 (3/9)

| Reference                                                                                                                                                           | Main contribution / evidence                                                        | Role in thesis                                                                             | Source and citation status    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------|
| R05   Platforms and profiling<br>Mittal & Vaishay (2019). A Survey of Techniques for Optimizing Deep Learning on GPUs. Journal of Systems Architecture, 99, 101635. | Classifies architecture- and system-level GPU optimization methods.                 | Provides a CUDA taxonomy for interpreting memory, precision, computation, and parallelism. | Peer-reviewed. DOCX and PPTX. |
| R06   Attention and memory efficiency<br>Dao et al. (2022). FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. NeurIPS.                   | Uses tiling and fusion to reduce HBM↔SRAM traffic while preserving exact attention. | Primary portable principle: reduce high-bandwidth-memory reads and writes.                 | Peer-reviewed. DOCX and PPTX. |

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R07-R08 (4/9)

| Reference                                                                                                                                              | Main contribution / evidence                                                         | Role in thesis                                                                 | Source and citation status    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------|
| R07   Attention and memory efficiency<br>Dao (2024). FlashAttention-2: Faster Attention with Better Parallelism and Work Partitioning. ICLR.           | Improves parallel work partitioning beyond the first FlashAttention kernel.          | Compare the IO-aware principle with backend-specific execution mapping.        | Peer-reviewed. DOCX and PPTX. |
| R08   Attention and memory efficiency<br>Shah et al. (2024). FlashAttention-3: Fast and Accurate Attention with Asynchrony and Low-precision. NeurIPS. | Uses Hopper-specific asynchronous pipelines, warp specialization, and low precision. | Key portability-limit example: later gains rely on NVIDIA-specific facilities. | Peer-reviewed. DOCX and PPTX. |

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Literature review evidence table — sources R09-R10 (5/9)

| Reference                                                                                                                                            | Main contribution / evidence                                                                          | Role in thesis                                                                | Source and citation status    |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------|
| R09   Attention and memory efficiency<br>Yuan et al. (2025). Native Sparse Attention: Hardware-Aligned and Natively Trainable Sparse Attention. ACL. | Combines hierarchical token compression, selection, and local windows with hardware-aligned sparsity. | For long context, measure quality and phase-specific latency, not only FLOPs. | Peer-reviewed. DOCX and PPTX. |
| R10   Attention and memory efficiency<br>Alizadeh et al. (2024). LLM in a Flash: Efficient Large Language Model Inference with Limited Memory. ACL.  | Selectively loads model parameters from flash storage into memory on demand.                          | Keep parameter-residency methods distinct from attention-kernel methods.      | Peer-reviewed. DOCX and PPTX. |

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Literature review evidence table — sources R11-R12 (6/9)

| Reference                                                                                                                                                       | Main contribution / evidence                                                    | Role in thesis                                                                                    | Source and citation status                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| R11   Serving<br>Kwon et al. (2023). Efficient Memory Management for Large Language Model Serving with PagedAttention. SOSP.                                    | Pages the KV cache to reduce fragmentation and improve sharing and batching.    | Measure isolated kernels and end-to-end serving separately; report cache and batching conditions. | Peer-reviewed. DOCX and PPTX.                                          |
| R12   Platforms and profiling<br>Benazir & Lin (2025). Profiling Large Language Model Inference on Apple Silicon: A Quantization Perspective. arXiv:2508.08531. | Profiles Apple Silicon and NVIDIA across model sizes and quantization settings. | Use system-level Metal profiling and state the precision and execution path explicitly.           | Preprint. DOCX and PPTX; local DOCX venue claim requires verification. |

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R13-R14 (7/9)

| Reference                                                                                                                                               | Main contribution / evidence                                                                                   | Role in thesis                                                                             | Source and citation status                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| R13   Platforms and profiling<br>Kirk & Hwu (2022). Programming<br>Massively Parallel Processors: A<br>Hands-on Approach (4th ed.).<br>Morgan Kaufmann. | Explains coalescing, tiling,<br>synchronization, occupancy, and CUDA<br>execution.                             | Use as background for CUDA diagnosis;<br>not direct evidence of Metal<br>portability.      | Textbook. PPTX only; citation<br>completed from standard<br>metadata.                  |
| R14   Platforms and profiling<br>Apple ML Research (2025). MLX /<br>Apple Silicon platform evidence<br>(exact report unspecified in local<br>files).    | Materials describe unified memory,<br>Metal Performance Primitives, and<br>distinct prompt/generation regimes. | Use only as contextual platform<br>evidence; keep it distinct from<br>independent studies. | Incomplete. DOCX and PPTX; add<br>exact report, author, date, URL,<br>and access date. |

Yellow rows need citation completion or verification before thesis submission.

## Literature review evidence table — sources R15-R16 (8/9)

| Reference                                                                                                                       | Main contribution / evidence                                                      | Role in thesis                                                                                       | Source and citation status                                                        |
|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| R15   Serving and deployment<br>Wu et al. (2019). Edge-inference deployment study (title and venue unspecified in local files). | Cited for operator support and hardware fragmentation in edge deployment.         | Document runtime/backend coverage; distinguish Metal GPU, Core ML, and ANE paths.                    | Incomplete. DOCX and PPTX; locate original title, venue, and identifier.          |
| R16   Serving and evaluation<br>MLCommons / MLPerf Inference benchmark suite (exact version unspecified in local files).        | Defines scenario-, quality-, version-, and run-rule-based performance evaluation. | Declare model, precision, shapes, versions, warm-up, repetitions, scenario, and quality constraints. | Incomplete. DOCX and PPTX; cite the precise MLPerf Inference version and ruleset. |

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Literature review evidence table — sources R17-R17 (9/9)

| Reference                                                                                                                                                             | Main contribution / evidence                                                 | Role in thesis                                                                            | Source and citation status    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------|
| R17   Serving and evaluation Schryen (2024). Speedup and Efficiency of Computational Parallelization: A Unifying Approach and Asymptotic Analysis. JPDC, 185, 104801. | Unifies speedup and efficiency models and analyzes overhead and scalability. | Report within-platform relative speedups and explain residual serial and system overhead. | Peer-reviewed. DOCX and PPTX. |

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# Source inventory and coverage

| Local thesis-directory file                    | Coverage decision                                                                                                                      |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Tursyn_CSCI693_Literature_Review_Introduction_ | Primary prose draft; selected-reference set.                                                                                           |
| Tursyn_CSCI693_Literature_Review_Introduction. | Near-duplicate prose draft; no additional identifiable sources.                                                                        |
| Tursyn_CSCI693_Literature_Review_Introduction2 | Earlier draft; adds incomplete references to Devlin, Apple MLX, Wu, and MLPerf.                                                        |
| Tursyn_CSCI693_Thesis_Literature_Review_Intro  | Presentation; adds Kirk & Hwu and confirms the mechanism-based review structure.                                                       |
| Sample Literature.pdf                          | Unrelated IoT/MEC-offloading excerpt. It has numbered citations but no supplied bibliography; excluded from this CUDA/Metal synthesis. |

**Coverage total: 17 individually identifiable CUDA/Metal works or sources (R01-R17).**

## Citation checks required before thesis submission

| Reference  | Issue                                                                              | Required action                                                         |
|------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| R12        | Local files name a PACM MACS venue; the identified 2025 work is an arXiv preprint. | Confirm a final peer-reviewed version, if any, before submission.       |
| R14        | Apple MLX/M5 platform evidence lacks title, author, publication date, and URL.     | Replace with the exact Apple report and an access date.                 |
| R15        | Wu et al. lacks title, venue, and persistent identifier.                           | Locate and enter a complete APA 7 citation before relying on its claim. |
| R16        | MLPerf Inference lacks a version, benchmark paper, and ruleset.                    | Cite the exact version and scenario used to guide the experiment.       |
| Sample PDF | The IoT/MEC sample has no reference pages in the supplied extract.                 | Obtain its bibliography before making a separate IoT-offloading table.  |